

Menglin Zhu

Curriculum Vitae with References

Updated: August 2, 2026

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Research Interest

- Probing quantum/energy materials atom by atom using advanced (S)TEM:
 - Multislice electron ptychography ▪ *Operando* microscopy ▪ 4D-STEM ▪ Spectroscopy
- Modeling atomistic behavior and phase transitions using DFT and *ab initio* MD simulations.
- Extracting insights from microscopy data using statistical analysis and machine learning

Education and Training

- 2023–present **Postdoctoral Researcher**, *Massachusetts Institute of Technology*, Prof. James M. LeBeau.
2018–2023 **Ph.D. in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.
2014–2017 **B.S. in Materials Science and Engineering**, *Ohio State University*, Prof. Jinwoo Hwang.

Honors & Awards

- 2026 **Albert Crewe Award** of Microscopy Society of America to recognize distinguished contributions to the field of microscopy in the physical sciences of an early career scientist
- 2025 **PARADIM user proposal #366** *Probing Dynamic Polar Structures in Relaxors*, 10 days of microscope time at Cornell PARADIM center
- 2023 **Postdoctoral Scholar** of Microscopy Society of America, awarded as a postdoctoral researcher for the paper submitted for the Microscopy and Microanalysis conference 2023
- 2022 **Student Scholar** of Microscopy Society of America, awarded as a Ph.D. candidate for the paper submitted for the Microscopy and Microanalysis conference 2022
- 2021 **Presidential Fellow** of Ohio State University, the most prestigious award given by the Graduate School to outstanding Ph.D. candidates
- 2017 **The Mars Fontana Scholarship** by Ohio State University, awarded to the most outstanding senior student in the Department of Material Science and Engineering
- 2016 **Summer Research Fellow** of Ohio State University
Markworth-Woolley Scholarship by Ohio State University

Publications

Citations: 1439, h-Index: 17, *Equal Contribution, #Correspondance

- [50] ²⁰²⁶ **Menglin Zhu***, Michael Xu*, Zishen Tian*, Colin Gilgenbach, Daniel Drury, Bridget R Denzer, Ching-Che Lin, Deokyoung Kang, Lane W Martin, and James M LeBeau. *Operando Electron Microscopy of Nanoscale Electronic Devices on Nonconductive Substrates. Microscopy and Microanalysis*, volume 32, page ozag007, DOI:10.1093/mam/ozag007.
- [49] ²⁰²⁶ **Menglin Zhu***, Michael Xu*, Yubo Qi, Colin Gilgenbach, Jieun Kim, Jiahao Zhang, Bridget R Denzer, Lane W Martin, Andrew M Rappe, and James M LeBeau. Bridging experiment and theory of relaxor ferroelectrics with multislice electron ptychography. *Science*, volume 392, pages 519–523, DOI:10.1126/science.ads6023.

- [48] ²⁰²⁶ Yu Yun, Liyan Wu, Drew Behrendt, Hao Pan, **Menglin Zhu**, Michael Xu, Anthony J Ruffino, John Carroll, Irina Baraban, Zishen Tian, Xianfei Xu, Dongfang Chen, Brendan M Hanrahan, Lane W Martin, James M LeBeau, Andrew M Rappe, Ilya Grinberg, and Jonathan E Spanier. Relaxor behavior in rocksalt cation-ordered material induced by (anti)ferroelectric phase competition. *Nat. Commun.*, pages 1–9, DOI:10.1038/s41467-026-74237-z.
- [47] ²⁰²⁶ Zishen Tian, **Menglin Zhu**, Jaegyu Kim, Piush Behera, Michael Xu, Thomas J Lee, Ching-Che Lin, Sreekeerthi Pamula, Archana Raja, Hao Pan, Jieun Kim, James M LeBeau, and Lane W Martin. Unleashing the electromechanical response of ferroelastic domain reorganization in mixed-phase tetragonal ferroelectric multilayers. *Adv. Mater.*, page e18417, DOI:10.1002/adma.202518417.
- [46] ²⁰²⁶ Colin Gilgenbach, **Menglin Zhu**, and James M LeBeau. Phaser: A unified and extensible framework for fast electron ptychography. *Npj Comput. Mater.*, pages 1–13, DOI:10.1038/s41524-026-01956-8.
- [45] ²⁰²⁵ **Menglin Zhu***, Michael Xu*, Yu Yun, Liyan Wu, Or Shafir, Colin Gilgenbach, Lane W Martin, Ilya Grinberg, Jonathan E Spanier, and James M LeBeau. Insights into chemical and structural order at planar defects in Pb₂MgWO₆ using multislice electron ptychography. *ACS Nano*, DOI:10.1021/acsnano.4c14833.
- [44] ²⁰²⁵ **Menglin Zhu***, Michael Xu*, Louis Alaerts, Hao Pan, Colin Gilgenbach, Geoffroy Hautier, Lane W Martin, and James M LeBeau. Direct determination of antiferroelectric-to-ferroelectric phase transition pathways in PbZrO₃ with Operando Electron Microscopy. *arXiv [cond-mat.mtrl-sci]*, DOI: arXiv2509.07194.
- [43] ²⁰²⁵ **Menglin Zhu***, Joseph Lanier*, Sevim Polat Genlik*, Jose G Flores, Victor da Cruz Pinha Barbosa, Mohit Randeria, Patrick M Woodward, Maryam Ghazisaeidi, Fengyuan Yang, and Jinwoo Hwang. Emergent ferromagnetism at LaFeO₃/SrTiO₃ interface arising from a strain-induced spin-state transition. *Adv. Mater. Interfaces*, volume 12, page 2500169, DOI:10.1002/admi.202500169.
- [42] ²⁰²⁵ Jith Sarker, Prachi Garg, Abrar Rauf, Ahsiaur Rahman Nirjhar, Hsien-Lien Huang, **Menglin Zhu**, A F M Anhar Uddin Bhuiyan, Lingyu Meng, Hongping Zhao, Jinwoo Hwang, Eric Osei-Agyemang, Saquib Ahmed, and Baishakhi Mazumder. Microscopic and spectroscopic investigation of (Al_xGa_{1-x})₂O₃ films: Unraveling the impact of growth orientation and aluminum content. *Adv. Mater. Interfaces*, volume 12, page 2301016, DOI:10.1002/admi.202301016.
- [41] ²⁰²⁵ Hao Pan, Liyan Wu, John Carroll, **Menglin Zhu**, Zishen Tian, Dongfang Chen, Hongrui Zhang, Xianzhe Chen, Xiaoxi Huang, Irina Baraban, Sreekeerthi Pamula, Cedric J G Meyers, R Ramesh, Kathleen Coleman, Brendan Hanrahan, James M LeBeau, Jonathan E Spanier, and Lane W Martin. Highly tunable relaxors developed from antiferroelectrics. *Adv. Mater.*, page e2505376, DOI:10.1002/adma.202505376.
- [40] ²⁰²⁵ Jieun Kim, Yubo Qi, Abinash Kumar, Yun-Long Tang, Michael Xu, Hiroyuki Takenaka, **Menglin Zhu**, Zishen Tian, Ramamoorthy Ramesh, James M LeBeau, Andrew M Rappe, and Lane W Martin. Size-driven phase evolution in ultrathin relaxor films. *Nat. Nanotechnol.*, pages 1–9, DOI:10.1038/s41565-025-01863-x.
- [39] ²⁰²⁵ I-Hsuan Kao, Junyu Tang, Gabriel Calderon Ortiz, **Menglin Zhu**, Sean Yuan, Rahul Rao, Jiahan Li, James H Edgar, Jiaqiang Yan, David G Mandrus, Kenji Watanabe, Takashi Taniguchi, Jinwoo Hwang, Ran Cheng, Jyoti Katoch, and Simranjeet Singh. Unconventional unidirectional magnetoresistance in heterostructures of a topological semimetal and a ferromagnet. *Nat. Mater.*, pages 1–9, DOI:10.1038/s41563-025-02175-0.
- [38] ²⁰²⁵ Deokyoung Kang, Xue-Zeng Lu, Megha Acharya, Sajid Husain, Isaac Harris, Piush Behera, Ching-Che Lin, Ella Banyas, Alex Smith, Francesco Ricci, **Menglin Zhu**, Bridget R Denzer, Tanguy Terlier, Shu Wang, Tae Yeon Kim, Lucas Caretta, Douglas Natelson, James M LeBeau, Jeffrey B Neaton, Ramamoorthy Ramesh, James M Rondinelli, and Lane W Martin. Bulk-Rashba effect with suppressed spin relaxation in a polar phase of Bi_{1-x}In_{1+x}O₃. *Adv. Mater.*, page e2504684, DOI:10.1002/adma.202504684.

- [37] ²⁰²⁵ Aaditya Bhat, Colin Gilgenbach, Junghwa Kim, Michael Xu, **Menglin Zhu**, and James M LeBeau. Sensitivity of multislice electron ptychography to point defects: A case study in SiC. *Ultramicroscopy*, volume 280, page 114282, DOI:10.1016/j.ultramic.2025.114282.
- [36] ²⁰²⁴ **Menglin Zhu**, Joseph Lanier, Jose Flores, Victor da Cruz Pinha Barbosa, Daniel Russell, Becky Haight, Patrick M Woodward, Fengyuan Yang, and Jinwoo Hwang. Structural degeneracy and formation of crystallographic domains in epitaxial LaFeO₃ films revealed by machine-learning assisted 4D-STEM. *Sci. Rep.*, volume 14, page 4198, DOI:10.1038/s41598-024-54661-1.
- [35] ²⁰²⁴ Sujan Shrestha, Yongseong Choi, Maximilian Krautloher, **Menglin Zhu**, Jinwoo Hwang, Bernhard Keimer, Ambrose Seo, and Jong-Woo Kim. Exploring magnetic anisotropy and robustness of the J_{eff} state under substantial orthorhombic distortion in Sr₂IrO₄ thin films. *Phys. Rev. B Condens. Matter*, volume 109, page 104415, DOI:10.1103/PhysRevB.109.104415.
- [34] ²⁰²⁴ Jith Sarker, Prachi Garg, Abrar Rauf, Ahsior Rahman Nirjhar, Hsien-Lien Huang, **Menglin Zhu**, A F M Anhar Uddin Bhuiyan, Lingyu Meng, Hongping Zhao, Jinwoo Hwang, Eric Osei-Agyemang, Saquib Ahmed, and Baishakhi Mazumder. Microscopic and spectroscopic investigation of (Al_xGa_{1-x})₂O₃ films: Unraveling the impact of growth orientation and aluminum content. *Adv. Mater. Interfaces*, DOI:10.1002/admi.202301016.
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- [28] ²⁰²³ Xiaolei Guo, Hsien-Lien Huang, **Menglin Zhu**, Karthikeyan Hariharan, Szu-Chia Chien, Ngan Huynh, Jinwoo Hwang, Wolfgang Windl, Christopher D Taylor, Eric J Schindelholz, and Gerald S Frankel. Interstitial elements created via metal 3D printing. *Mater. Today*, volume 66, pages 92–104, DOI:10.1016/j.mattod.2023.04.020.
- [27] ²⁰²² **Menglin Zhu**, and Jinwoo Hwang. Scattering angle dependence of temperature susceptibility of electron scattering in scanning transmission electron microscopy. *Ultramicroscopy*, volume 232, page 113419, DOI:10.1016/j.ultramic.2021.113419.

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- [23]₂₀₂₂ Olivia G Licata, **Menglin Zhu**, Jinwoo Hwang, and Baishakhi Mazumder. Nanoscale chemistry and ion segregation in zirconia-based ceramic at grain boundaries by atom probe tomography. *Scr. Mater.*, volume 213, page 114603, DOI:10.1016/J.SCRIPTAMAT.2022.114603.
- [22]₂₀₂₂ Hyunseok Kim, Yunpeng Liu, Kuangye Lu, Celesta S Chang, Kuan Qiao, Ki Seok Kim, Bo-In Park, Junseok Jeong, **Menglin Zhu**, Jun Min Suh, Yongmin Baek, You Jin Ji, Sungsu Kang, Sangho Lee, Ne Myo Han, Chansoo Kim, Chanyeol Choi, Xinyuan Zhang, Haozhe Wang, Lingping Kong, Jungwon Park, Kyusang Lee, Geun Young Yeom, Sungkyu Kim, Jinwoo Hwang, Jing Kong, Sang-Hoon Bae, Wei Kong, and Jeehwan Kim. Multiplication of freestanding semiconductor membranes from a single wafer by advanced remote epitaxy, DOI:10.48550/arxiv.2204.08002.
- [21]₂₀₂₂ Hyunseok Kim, Sangho Lee, Jiho Shin, **Menglin Zhu**, Marx Akl, Kuangye Lu, Ne Myo Han, Yongmin Baek, Celesta S Chang, Jun Min Suh, Ki Seok Kim, Bo In Park, Yanming Zhang, Chanyeol Choi, Heechang Shin, He Yu, Yuan Meng, Seung Il Kim, Seungju Seo, Kyusang Lee, Hyun S Kum, Jae Hyun Lee, Jong Hyun Ahn, Sang Hoon Bae, Jinwoo Hwang, Yunfeng Shi, and Jeehwan Kim. Graphene nanopattern as a universal epitaxy platform for single-crystal membrane production and defect reduction. *Nat. Nanotechnol.*, volume 17, pages 1054–1059, DOI:10.1038/s41565-022-01200-6.
- [20]₂₀₂₂ Md Rezaul Karim, Benthara Hewage Dinushi Jayatunga, Kaitian Zhang, **Menglin Zhu**, Jinwoo Hwang, Kathleen Kash, and Hongping Zhao. Band Structure Engineering Based on InGa_N/ZnGeN₂ Heterostructure Quantum Wells for Visible Light Emitters. *Cryst. Growth Des.*, volume 22, pages 131–139, DOI:10.1021/acs.cgd.1c00630.
- [19]₂₀₂₂ I Hsuan Kao, Ryan Muzzio, Hantao Zhang, **Menglin Zhu**, Jacob Gobbo, Sean Yuan, Daniel Weber, Rahul Rao, Jiahua Li, James H Edgar, Joshua E Goldberger, Jiaqiang Yan, David G Mandrus, Jinwoo Hwang, Ran Cheng, Jyoti Katoch, and Simranjeet Singh. Deterministic switching of a perpendicularly polarized magnet using unconventional spin–orbit torques in WTe₂. *Nature Materials* 2022 21:9, volume 21, pages 1029–1034, DOI:10.1038/s41563-022-01275-5.
- [18]₂₀₂₁ Tiancong Zhu, Alexander J Bishop, Tong Zhou, **Menglin Zhu**, Dante J O'Hara, Alexander A Baker, Shuyu Cheng, Robert C Walko, Jacob J Repicky, Tao Liu, and Others. Synthesis, Magnetic Properties, and Electronic Structure of Magnetic Topological Insulator MnBi₂Se₄. *Nano Lett.*, DOI:10.1021/acs.nanolett.1c00141.
- [17]₂₀₂₁ Md Rezaul Karim, Brenton A Noesges, Benthara Hewage Dinushi Jayatunga, **Menglin Zhu**, Jinwoo Hwang, Walter R L Lambrecht, Leonard J Brillson, Kathleen Kash, and Hongping Zhao. Experimental determination of the valence band offsets of ZnGeN₂ and (ZnGe)_{0.94}Ga_{0.12}N₂ with GaN. *J. Phys. D Appl. Phys.*, volume 54, page 245102, DOI:10.1088/1361-6463/abee45.

- [16]₂₀₂₁ Soohyun Im, Yunzhi Yuchi Wang, Pengyang Zhao, Geun Hee Yoo, Zhen Chen, Gabriel Calderon, Mehrdad Abbasi Gharacheh, **Menglin Zhu**, Olivia Licata, Baishakhi Mazumder, David A Muller, Eun Soo Park, Yunzhi Yuchi Wang, and Jinwoo Hwang. Medium-range ordering, structural heterogeneity, and their influence on properties of Zr-Cu-Co-Al metallic glasses. *Physical Review Materials*, volume 5, pages 1–11, DOI:10.1103/PhysRevMaterials.5.115604.
- [15]₂₀₂₁ Yang Cheng, Sisheng Yu, **Menglin Zhu**, Jinwoo Hwang, and Fengyuan Yang. Tunable topological Hall effects in noncollinear antiferromagnet Mn₃Sn/Pt bilayers. *APL Materials*, volume 9, page 51121, DOI:10.1063/5.0048733.
- [14]₂₀₂₀ Tiancong Zhu, Dante J O'Hara, Brenton A Noesges, **Menglin Zhu**, Jacob J Repicky, Mark R Brenner, Leonard J Brillson, Jinwoo Hwang, Jay A Gupta, and Roland K Kawakami. Coherent growth and characterization of van der Waals 1T-VSe₂ layers on GaAs (111) B using molecular beam epitaxy. *Physical Review Materials*, volume 4, page 84002, DOI:10.1103/PhysRevMaterials.4.084002.
- [13]₂₀₂₀ Sujan Shrestha, Matthew Coile, **Menglin Zhu**, Maryam Souri, Jiwoong Kim, Rina Pandey, Joseph W Brill, Jinwoo Hwang, Jong-Woo Kim, and Ambrose Seo. Nanometer-Thick Sr₂IrO₄ freestanding films for flexible electronics. *ACS Applied Nano Materials*, volume 3, pages 6310–6315, DOI:10.1021/acsanm.0c01351.
- [12]₂₀₂₀ Taeseon Lee, **Menglin Zhu**, Taylor Dittrich, Jinwoo Hwang, Anupam Vivek, and Glenn S Daehn. Microstructural Investigation of the Impact Weld Interface of Pseudo Single Grained Cu and Ag. *Metall. Mater. Trans. A*, volume 51, pages 558–561, DOI:10.1007/s11661-019-05557-7.
- [11]₂₀₂₀ Aidan J Lee, Adam S Ahmed, Brendan A McCullian, Side Guo, **Menglin Zhu**, Sisheng Yu, Patrick M Woodward, Jinwoo Hwang, P Chris Hammel, and Fengyuan Yang. Interfacial Rashba-Effect-Induced Anisotropy in Nonmagnetic-Material–Ferrimagnetic-Insulator Bilayers. *Phys. Rev. Lett.*, volume 124, page 257202, DOI:10.1103/PhysRevLett.124.257202.
- [10]₂₀₂₀ Md Rezaul Karim, Benthara Hewage Dinushi Jayatunga, **Menglin Zhu**, Rebecca A Lalk, Olivia Licata, Baishakhi Mazumder, Jinwoo Hwang, Kathleen Kash, and Hongping Zhao. Effects of cation stoichiometry on surface morphology and crystallinity of ZnGeN₂ films grown on GaN by metalorganic chemical vapor deposition. *AIP Adv.*, volume 10, page 65302, DOI:10.1063/1.5137767.
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- [8]₂₀₂₀ Yang Cheng, Sisheng Yu, **Menglin Zhu**, Jinwoo Hwang, and Fengyuan Yang. Electrical Switching of Tristate Antiferromagnetic Néel Order in α -Fe₂O₃ Epitaxial Films. *Phys. Rev. Lett.*, volume 124, page 27202, DOI:10.1103/PhysRevLett.124.027202.
- [7]₂₀₂₀ A F M Anhar Uddin Bhuiyan, Zixuan Feng, Jared M Johnson, Hsien-Lien Huang, Jith Sarker, **Menglin Zhu**, Md Rezaul Karim, Baishakhi Mazumder, Jinwoo Hwang, and Hongping Zhao. Response to “Comment on ‘Phase transformation in MOCVD growth of (Al_xGa_{1-x})₂O₃ thin films””[APL Mater. 8, 089101 (2020)]. *APL Materials*, volume 8, page 89102, DOI:10.1063/5.0014806.
- [6]₂₀₂₀ A F M Anhar Uddin Bhuiyan, Zixuan Feng, Jared M Johnson, Hsien-Lien Huang, Jith Sarker, **Menglin Zhu**, Md Rezaul Karim, Baishakhi Mazumder, Jinwoo Hwang, and Hongping Zhao. Phase transformation in MOCVD growth of (Al_xGa_{1-x})₂O₃ thin films. *APL Materials*, volume 8, page 31104, DOI:10.1063/1.5140345.
- [5]₂₀₁₉ Baishakhi Mazumder, Jith Sarker, Yuewei Zhang, Jared M Johnson, **Menglin Zhu**, Siddharth Rajan, and Jinwoo Hwang. Atomic scale investigation of chemical heterogeneity in β -(Al_xGa_{1-x})₂O₃ films using atom probe tomography. *Appl. Phys. Lett.*, volume 115, page 132105, DOI:10.1063/1.5113627.

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Teaching Experience

- Fall, 2023 **3.34 Imaging of Materials**, guest lecturer, Massachusetts Institute of Technology
- Fall, 2020 **MSE3151 Transport Phenomenon and Kinetics**, teaching assistant, Ohio State University
- Fall, 2019 **MSE3332 Undergraduate Lab II**, laboratory assistant and instructor, Ohio State University

Synergistic Activities

- 2023-present **Mentor of two graduate students** on project *Collaborative for Hierarchical Agile and Responsive Materials (CHARM) under cooperative agreement W911NF-19-2-011*
- 2023 **Co-organize Microscopy and Microanalysis conference P08 symposium** entitled *Atomic Scale Microscopy of Interfaces and Heterostructures with Correlated Phenomena*
- 2023 **Mentor of two REU Students** on project *NFO Thin Films Grown via an Off-Axis Sputtering Method; STEM Characterization of NdFeO₃/SrTiO₃ Thin Films*
- 2022 **Mentor of one REU Student** on project *Effects of Octahedral Tilting and Lattice Strain on LaFeO₃*

References

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